

CompTIA A+ Assessment AP-A1

Hardware Components & System Assembly — Submission Answers

Program: CompTIA A+ | Stack: IT Support | Time: 3 Hours

Note on the practical tasks: Sections B and C ask for photos/screenshots of real hardware work. Where physical hardware is not available, the instructions allow clearly labelled written documentation and diagrams, which is what is provided here. The example build used throughout is: ASUS PRIME B660M-A D4 (LGA1700, DDR4), Intel Core i5-12400, 16 GB DDR4-3200 (2 x 8 GB), 512 GB NVMe M.2 SSD, a PCIe x16 GPU, 550 W 80+ PSU, stock cooler. Replace these with the exact make/model of the system you actually handle, and attach your own photos where marked [PHOTO].

Section A — Concept Application

S1. Selecting compatible motherboard, CPU, RAM and storage

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

Process. I start from the workload (office and productivity applications, multitasking, web, video calls) and a budget, then choose components in a fixed order because each choice constrains the next:

- **1. CPU first.** For office use a mainstream 4-6 core CPU with integrated graphics is enough (e.g., Intel Core i5-12400 or AMD Ryzen 5 with iGPU), which also removes the need for a discrete GPU. This fixes the **socket** (LGA1700 for Intel 12th-14th gen, AM4/AM5 for Ryzen) and the **memory type** the CPU supports.
- **2. Motherboard.** Choose a board with the same socket and a chipset that supports the chosen CPU (e.g., B660/H610 for LGA1700). Check the form factor against the case (ATX, micro-ATX, mini-ITX), the number of RAM slots, M.2/SATA ports, USB and display outputs, and that the BIOS version supports the CPU generation (the manufacturer's CPU support list shows the minimum BIOS version; some boards need a BIOS-flashback update first).
- **3. RAM.** Match the generation the board supports (DDR4 or DDR5; they are physically keyed differently and not interchangeable), a speed within the board/CPU supported range, and capacity (16 GB is a good office baseline; 8 GB is minimum). Use a matched kit in dual-channel (2 x 8 GB) for better bandwidth, and confirm the form factor (DIMM for desktop, SO-DIMM for laptops).
- **4. Storage.** Choose an SSD. If the board has an M.2 slot, confirm it supports NVMe (PCIe) vs SATA M.2 and the PCIe generation; otherwise use a 2.5-inch SATA SSD with a SATA cable. Check the slot's keying (M-key) and length (2280).

How I confirm they work together. (a) Read the motherboard QVL (Qualified Vendor List) and CPU support list on the manufacturer's site; (b) use a tool such as PCPartPicker, which flags socket, memory-type and form-factor conflicts; (c) compare data sheets: socket, chipset, DDR generation, max RAM speed/capacity, M.2 interface and PCIe lanes; (d) confirm the PSU and cooler support the socket and have enough connectors; (e) after assembly, verify in BIOS/UEFI that the CPU model, total RAM, and storage drive are all detected, then run a short stress test.

Documented checks before purchase prevent returns and boot failures.

S2. HDD vs SSD vs NVMe for an office workload

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Feature	HDD	SATA SSD	NVMe SSD
Technology	Spinning magnetic platters with moving read/write heads	NAND flash, no moving parts	NAND flash, no moving parts
Interface / protocol	SATA III (AHCI), 6 Gbps	SATA III (AHCI), 6 Gbps	PCIe (NVMe protocol), usually M.2, 4 lanes
Feature	HDD	SATA SSD	NVMe SSD
Typical sequential speed	80-160 MB/s	~500-550 MB/s	~2,000-7,000+ MB/s (Gen3/Gen4/Gen5)
Access time / latency	~5-10 ms (seek + rotation)	~0.1 ms	Lower still, with deep command queues
Durability / noise	Fragile when moving, audible, vibration	Shock resistant, silent	Shock resistant, silent; may need heatsink
Power / heat	Higher, motor-driven	Low	Low idle, can run hot under sustained load
Cost per GB	Lowest	Medium	Medium-high (gap is narrowing)
Form factor	3.5" / 2.5"	2.5" or M.2 SATA	M.2 2280 (also U.2, PCIe card)

Recommendation: SATA or NVMe SSD, with a mainstream NVMe SSD (256-512 GB) preferred where the motherboard has an M.2 slot. Office workloads are made of many small random reads/writes (OS boot, launching applications, opening documents, browser caches), where the dominant factor is access latency and IOPS, not raw sequential throughput. An SSD improves these by an order of magnitude over an HDD: boot in seconds, applications open almost instantly, no seek delays. SSDs have no moving parts so they are more reliable in busy offices, use less power, run silently, and survive being moved. A mainstream NVMe drive costs about the same as a SATA SSD today, needs no data or power cables (cleaner build, better airflow), and gives headroom for future use. A top-end Gen4/Gen5 NVMe is not necessary for this workload; the extra bandwidth would go unused.

An HDD is only appropriate as secondary bulk storage for backups/archives.

S3. GPU and cooling under sustained graphics load

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

GPU. The graphics processing unit is a massively parallel processor with its own dedicated memory (VRAM) that handles rendering, display output, and acceleration of design applications (viewport rendering, 2D/3D effects, filters, GPU-accelerated compute). Offloading these tasks from the CPU keeps the interface smooth and leaves the CPU free for other work. For mildly graphics-intensive design software, a mid-range discrete GPU (or a strong integrated GPU) with enough VRAM (typically 4-8 GB), correct drivers, and sufficient PCIe x16 bandwidth is appropriate. Certified/up-to-date drivers matter for stability.

Cooling. CPU and GPU convert electrical power to heat. If heat is not removed, silicon temperatures rise until **thermal throttling** reduces clock speeds (lower performance), and in the worst case the system shuts down or components degrade. For several hours of daily load the cooling system must remove heat continuously:

- **CPU cooler** (heatsink + fan or AIO liquid) with **thermal paste** correctly applied to fill microscopic gaps between the CPU heat spreader and cooler base.
- **GPU cooler** (heatsink, heat pipes, fans) which exhausts heat from the card.
- **Case airflow:** front/bottom intake fans, rear/top exhaust, positive or balanced pressure, clean dust filters, tidy cable management, and adequate room around the case.
- **Monitoring:** use BIOS/UEFI hardware monitor or software (HWMonitor, GPU-Z) to check temperatures and fan RPM under load; healthy loaded temperatures are generally below about 80-85 degrees C.

Together, an adequately specified GPU and effective cooling sustain boost clocks, prevent throttling and crashes, extend component life, and keep fan noise reasonable.

S4. Why hardware compatibility matters

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Components from different vendors only work if they share the same physical standards, electrical interfaces and firmware support. Incompatibility wastes money and time, can prevent boot, and in some cases can damage parts (for example, forcing a wrong CPU into a socket bends pins). Compatibility therefore must be verified before purchase and before installation.

Incompatibility	Resulting problem
CPU socket / chipset mismatch (e.g., LGA1700 CPU on AM5 board)	CPU cannot physically fit; or chipset does not support the CPU generation
BIOS too old for the CPU	No POST / no display until BIOS is updated
Wrong RAM type (DDR4 in DDR5 board) or unsupported speed/capacity	Will not seat (different key notch); or fails to boot, runs at lower speed, or is unstable
Form factor mismatch (ATX board in micro-ATX case; wrong standoffs)	Board does not fit or can short against the case
Storage interface mismatch (NVMe in a SATA-only M.2 slot, or M.2 length)	Drive not detected in BIOS
PSU too weak or missing connectors (e.g., 8-pin CPU, PCIe power)	Random shutdowns, no power-on, or GPU/CPU cannot be powered
Cooler does not support socket, or GPU too long/thick for case	Overheating, throttling, or cannot be installed
OS / driver unsupported	Hardware not recognised or unstable

S5. Choosing a PSU and pre-power safety/connection checks

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

Choosing the PSU.

- **Wattage:** add up the peak (TDP/TBP) power of the CPU, GPU, and other components (drives, fans, RAM) and add 20-30% headroom. Online PSU calculators help. Example: CPU ~65 W + GPU ~150 W + rest ~50 W = ~265 W, so a quality 500-550 W unit is suitable.
- **Connectors:** 24-pin ATX motherboard, 4+4-pin (EPS) CPU power, PCIe 6/8-pin for the GPU, SATA power for SATA drives; check the GPU's required connectors.
- **Efficiency and quality:** 80 PLUS Bronze or better, from a reputable brand, with protections (OVP, UVP, OCP, SCP, OTP). Check the 12V rail rating, not just the total wattage.
- **Form factor:** ATX or SFX matching the case; modular or semi-modular for tidier cabling. Voltage switch set correctly for the region (230 V in India, where applicable).

Safety and connection checks before power-on.

- Wear an ESD wrist strap or touch bare case metal; work on a non-conductive, uncluttered surface; PSU switch set to 0/off and power cord unplugged while connecting.
- Check the 24-pin ATX and CPU 4/8-pin EPS connectors are fully clicked in, with the latch engaged (do not confuse the CPU EPS plug with a PCIe plug).
- Connect PCIe power to the GPU (correct number of pins, no daisy-chain overload) and SATA power/data to drives; M.2 NVMe drives are screwed down.
- Connect case front-panel headers (power switch, reset, LEDs, USB, audio) per the motherboard manual.
- Confirm RAM and GPU are fully seated, CPU cooler is mounted with paste and its fan header is connected (CPU_FAN), and no loose screws or standoffs short the board.
- Route cables away from fans; make sure no cable is pinched and no connector is left unplugged.
- Connect monitor to the correct output (GPU, not motherboard, if a discrete card is installed), plug in the power cord, switch the PSU on and press the power button; watch/listen for fans, lights, beeps.

S6. POST verification and what to do when POST fails

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

POST (Power-On Self-Test) is the firmware's hardware check that runs before the operating system loads. Successful POST indicators:

- Fans, drives and LEDs start; the keyboard lights; the manufacturer logo/POST screen appears on the monitor.
- A single short beep (on boards with a speaker) or no error codes on the motherboard debug LED/Q-code display.
- Entering BIOS/UEFI (Del/F2) shows the correct CPU model, total RAM, storage drive(s), and sane temperatures and fan RPM.
- The system proceeds to the boot device/OS installer without error messages.

If POST fails:

- Power off, unplug, and observe the beep codes / debug LEDs / error text, then look them up in the board manual (CPU, DRAM, VGA, BOOT).
- Check power first: PSU switch, 24-pin and CPU EPS connectors, and the front-panel power header.
- Reseat RAM (try one stick in the slot recommended by the manual), then CPU and GPU; make sure the monitor cable is plugged into the correct output.
- Disconnect everything non-essential (extra drives, cards, USB devices) and boot with the minimum: CPU, one RAM stick, PSU (and GPU if no iGPU).
- Clear CMOS (jumper or battery removal) and update BIOS if the CPU is new for the board.
- Swap with known-good parts (RAM, PSU) to isolate the faulty component, replace it, and document the fix.

Section B — Practical Tasks (Written Documentation)

Task 1: Component Identification & Functions

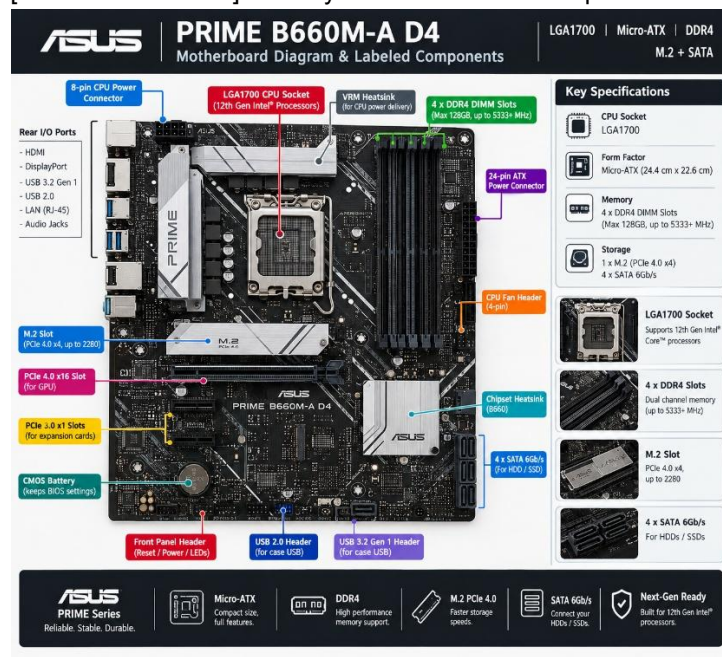
Component	Function	Make / key specification (example replace with your own)
Motherboard	Main circuit board that connects and lets all components communicate; provides the CPU socket, RAM slots, expansion slots, storage ports, I/O, chipset and BIOS/UEFI firmware.	ASUS PRIME B660M-A D4; LGA1700, micro-ATX, 4 DDR4 slots, M.2 + SATA
CPU	The 'brain': executes instructions and performs calculations for the OS and applications.	Intel Core i5-12400; 6 cores / 12 threads, up to 4.4 GHz
RAM	Fast, volatile working memory that holds data and programs currently in use.	16 GB (2 x 8 GB) DDR4-3200 DIMM
GPU	Processes graphics and display output; accelerates rendering with its own VRAM.	Discrete PCIe x16 card, 4-8 GB GDDR (or CPU's integrated graphics)
Storage device	Non-volatile storage for the OS, applications and user data.	512 GB NVMe M.2 SSD (PCIe), M.2 2280
Cooling system	Removes heat from CPU/GPU/case to prevent throttling and damage.	CPU heatsink + fan with thermal paste; case fans (intake/exhaust)
PSU (supporting)	Converts AC mains to regulated DC (+12 V, +5 V, +3.3 V) for components.	550 W, 80 PLUS Bronze, ATX

Task 2: Motherboard Ports & Socket Identification

Item	Identification / purpose
CPU socket	LGA1700 (Intel). Compatible with 12th/13th/14th-gen Core (Alder/Raptor Lake) CPUs, subject to chipset/BIOS support. (For AMD examples: AM4 = Ryzen 1000-5000, AM5 = Ryzen 7000+.)
24-pin ATX power	Main power from PSU to the board (right edge).
4/8-pin EPS (CPU power)	Dedicated 12 V supply to the CPU (top-left near the socket).
DIMM slots (4)	DDR4 RAM; fill A2/B2 first for dual-channel.
SATA ports (4-6)	Connect 2.5"/3.5" drives and optical drives via SATA cables.
M.2 slots	M.2 2280 for NVMe/SATA SSDs (direct to CPU/chipset PCIe lanes).
PCIe x16 slot	Primary GPU slot (CPU lanes, Gen4/Gen5 depending on board).
PCIe x1 / x4 slots	Add-in cards: network, sound, capture, USB, storage controllers.
USB headers (USB 2.0 / 3.0 / Type-C)	Connect front-panel USB ports of the case.
Front panel header (F_PANEL)	Power switch, reset switch, HDD and power LEDs.
CPU_FAN / CHA_FAN headers	4-pin PWM fan power/control.
Audio header (HD_AUDIO)	Front-panel audio jacks.

CMOS battery (CR2032)	Keeps BIOS settings and clock.
Rear I/O	USB ports, HDMI/DisplayPort/VGA (iGPU output), LAN (RJ-45), audio jacks, PS/2 on some boards.
BIOS/UEFI chip, CLR_CMOS jumper	Firmware storage; jumper clears settings.

[PHOTO/DIAGRAM] Attach your own motherboard photo with each of the above labelled.



Task 3: RAM & Storage Installation

RAM removal and reinstallation

- **ESD precautions:** PC shut down, PSU switched off and unplugged, hold power button to drain residual power; wear an anti-static wrist strap clipped to bare metal of the case (or touch the chassis frequently); work on an anti-static mat; handle the module by its edges only.
- **Remove:** push both retaining clips at the ends of the DIMM slot outward; the module lifts slightly; pull it straight out by its edges and place it in an anti-static bag.
- **Reinstall:** align the notch in the module with the key in the slot (correct orientation), place it straight over the slot, press down firmly on both ends until the clips snap back automatically.
- **Seating check:** clips fully closed; module is level and the gold contacts are not visible; no gap at either end.

Storage device removal and reinstallation (NVMe M.2 used in this example)

- **Connection type:** NVMe M.2 PCIe (M-key, 2280), no cable needed. (For a SATA SSD: SATA data cable to a motherboard SATA port + SATA power cable from the PSU.)
- **Remove:** remove the heatsink if fitted, unscrew the single retaining screw so the drive pops up at an angle, then pull it out of the slot by the edges.
- **Reinstall:** insert the drive at roughly 30 degrees into the slot until the contacts are fully inserted, press it flat and secure it with the screw at the standoff (not over-tightened), refit the heatsink.

Detection check after reinstallation

- Power on and enter BIOS/UEFI: the total memory (e.g., 16384 MB) matches the installed capacity and speed; the NVMe drive is listed under Storage Information / boot options.
- In the OS: Task Manager > Performance (Memory/Disk), Disk Management or *System Information* show full RAM and the drive. Optional: run Windows Memory Diagnostic.

Task 4: Full Desktop PC Assembly

Step	Action	ESD / safety note
1	Prepare workspace; gather tools (Phillips #2 screwdriver), manuals, anti-static strap; inspect the case and parts for damage.	Wear ESD strap; PSU unplugged.
2	Install CPU on the motherboard (outside the case): lift the retention lever, align the triangle marker/notches, lower the CPU in without force, close the lever and plate.	Hold CPU by edges; never touch socket pins/contacts.
3	Install the CPU cooler: apply a pea-sized amount of thermal paste (if not pre-applied), mount the cooler, tighten diagonally, connect the CPU_FAN header.	Remove protective plastic film from the cooler.
4	Install RAM into slots A2 and B2 for dual-channel until the clips click.	Edges only.
5	Install the M.2 NVMe SSD into the M.2 slot and screw it down.	Remove protective sticker from the thermal pad.
6	Install the I/O shield and motherboard standoffs; place the board in the case and secure with screws.	Standoffs must match the board holes only.
7	Install the PSU and cable the 24-pin, 4+4-pin EPS, and SATA power if needed.	PSU off and unplugged.
8	Install the GPU in the top PCIe x16 slot (remove slot covers); secure screws; connect PCIe power cables.	Latch must click; use all required power connectors.
9	Connect front panel headers (power/reset/LED), USB, audio, and case fans.	Check the manual pin diagram.
10	Cable management; close-check: nothing loose, no cables near fans.	No metal debris left inside.
11	Connect monitor (to the GPU output), keyboard, mouse; plug in power; switch the PSU on; press the power button.	Keep case panels off for first test.
12	Observe POST: fans spin, logo/POST screen shows; enter BIOS and verify CPU, RAM and drive are detected and temperatures are normal.	If no POST: follow troubleshooting in Section A S6.

PC BUILD GUIDE

From parts to power - step by step.

ESD & SAFETY PRECAUTIONS

- Work on a clean, static-free surface.
- Wear an anti-static wrist strap to bond a grounded metal surface.
- Keep components in anti-static bags until use.
- Power off and unplug the PSU before installation.
- Handle components by the edges, avoid touching pins or contacts.

PARTS LIST

Motherboard ASUS PRIME B560M-A D4 (LGA1700, mATX)	CPU Intel Core i5-12400F (LGA1700)	RAM 2 x 8GB DDR4 (DDR4000)	Storage Western Digital 1TB M.2	GPU NVIDIA GeForce RTX 3060 (Dual Fan)	PSU 650W 80+ Bronze (ATX)
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1 Install CPU

- Lift the CPU socket lever.
- Align the CPU (match the triangle).
- Gently place it and close the lever.
- Apply thermal paste (if not pre-applied).

2 Install RAM

- Open the slot clips.
- Insert the RAM into the PCIe x16 slot.
- Press down until clips lock.

3 Install M.2 SSD

- Locate the M.2 slot (on motherboard).
- Insert the SSD at an angle and push down.
- Secure with the screw.

4 Install Motherboard in Case

- Align the standoffs with the motherboard holes.
- Place the board and secure with screws.
- Ensure correct I/O shield is in place.

5 Install PSU

- Place the PSU in the case (vent down if needed).
- Secure with screws.
- Connect main power cables (24-pin, 8-pin EPS).

6 Install GPU

- Remove PCIe slot covers.
- Insert the GPU into the PCIe x16 slot.
- Secure with screws.
- Connect PCIe power cables (8-pin).

7 Connect Case Cables

- Connect front panel (power, reset, LEDs).
- Connect USB and I/O audio cables.
- Organize cables for better airflow.

8 Final Check Before Power On

- Check all connections (24-pin, 8-pin, PCIe).
- Ensure no loose screws or cables.
- Verify RAM, GPU, and SSD are properly seated.

9 Power On

- Connect monitor, keyboard and mouse.
- Plug in the power cables.
- Turn on the PSU switch and press the case power button.

10 POST Screen

- The system should show the motherboard logo.
- It will proceed to POST (Power-On Self Test).
- Wait for the OS boot screen (or BIOS).

11 BIOS Confirmation

- Press Del or F2 to enter BIOS (if needed).
- Check CPU, RAM, and storage are detected.
- Save & exit (F10).

12 Done!

- Your PC is built and running successfully!
- Ready for OS installation and setup.

BUILD COMPLETE!

CPU + RAM + STORAGE + GPU + PSU = YOUR PC

TIPS:

- Keep drivers and BIOS updated.
- Monitor temps during first boot.
- Enjoy your new build!

ASUS PRIME B560M-A D4 (LGA1700 + mATX + DDR4)

Build • Play • Perform

Section C — Mini Capstone: Workstation Deployment & Boot-Failure Diagnosis

System assembled/inspected (example): ASUS PRIME B660M-A D4 motherboard (LGA1700, DDR4), Intel Core i5-12400, 2 x 8 GB DDR4-3200, 512 GB NVMe SSD, 550 W PSU. The report below uses this system as reference.

1. Three components to check first when a new PC fails to boot

#	Component	Why first
1	PSU and power connections (24-pin, CPU EPS, PSU switch, wall power, front-panel power header)	No power = no fans or lights. It is the most common cause of 'dead' systems and the easiest to check; an unplugged EPS or 24-pin connector, a switched-off PSU, or a miswired power header stops the system from starting at all.
2	RAM (seating, slot, compatibility)	Poorly seated, wrong-type or faulty RAM is the most frequent cause of no POST/no display and beep codes (long repeating beeps, DRAM debug LED). It is quick to reseat.
3	CPU / motherboard (CPU seated correctly, cooler fitted, no bent pins, BIOS supports the CPU)	If the CPU is misaligned, the BIOS is too old for the CPU, or the cooler/standoffs are wrong, the system fails to POST. Checked after power and RAM because it is more invasive.

Storage and GPU are checked next: a missing drive normally gives a 'no boot device' message after POST, while a loose GPU gives no video output.

2. Motherboard-CPU compatibility and system functionality

The CPU and motherboard must match in **socket**, **chipset**, **BIOS version** and **memory support**. In the reference system, the i5-12400 is an LGA1700 CPU and the PRIME B660M-A D4 has an LGA1700 socket with a B660 chipset, so the CPU fits physically and electrically. The B660 chipset supports 12th-gen CPUs, but 13th/14th-gen CPUs need a BIOS update; installing such a CPU on an old BIOS would result in no POST. The CPU's integrated memory controller determines memory support; this board is the DDR4 version, so DDR5 modules cannot be used. The chipset also determines the number of PCIe lanes, USB ports and M.2 slots. If any of these do not match, the symptoms in the ABC Solutions lab (dead systems, no display, detection errors) will appear. Therefore verifying the socket/chipset/BIOS in the manufacturer's CPU support list before assembly avoids those failures.

3. Verifying RAM and storage installation (BIOS/POST)

- **Physical:** RAM clips locked, module level, in the recommended slots; M.2 drive fully inserted and screwed down or SATA data and power cables firmly connected at both ends.
- **POST:** the system must complete POST without memory beeps, the DRAM debug LED, or error messages (e.g., 'No boot device found').
- **BIOS/UEFI:** the main page should show the total installed memory (16 GB) and speed (XMP/DOCP profile if enabled); the Storage/Boot menu should list the SSD with the correct model and capacity.
- **OS level:** Windows Task Manager/System Information shows full RAM; Disk Management shows the drive. If not detected, reseat, try another slot/port, and test a known-good module/driver.

4. Hardware-installation checklist for future deployments

#	Checklist step	Done
1	Before purchase: verify CPU socket/chipset/BIOS, RAM type and speed, M.2/SATA support, case and cooler clearances using the board manual and QVL.	
2	Before touching parts: use ESD strap/mat, unplug the PSU, work on a clean flat surface.	

3	Install the CPU, cooler (with thermal paste), RAM and M.2 drive on the board; confirm the CPU is aligned and RAM clips click.	
#	Checklist step	Done
4	Mount the board with correct standoffs; connect the 24-pin, CPU EPS, GPU power, SATA power/data and front-panel headers; check every connector latch.	
5	First power-on with panels open: confirm fans spin, POST completes, and BIOS shows the correct CPU, RAM and drive; update BIOS if necessary.	
6	Run a short test (memory test, CPU/GPU stress) and check temperatures; then close the case, tidy cables, label, and record serial numbers/results in the deployment log.	

5. Technical justification for SSDs over HDDs in office workstations

The lab workstations run operating systems, office suites, browsers and email, which generate many small random reads/writes. HDDs are limited by mechanical seek time (5-10 ms) and rotational latency, so boot and application launch are slow and multitasking stalls. SSDs use NAND flash with sub-millisecond access time and far higher IOPS, so a PC boots in seconds, applications launch almost immediately and user productivity increases across dozens of machines. SSDs have no moving parts, which gives better resistance to knocks, lower failure rates, no noise, and lower power and heat output (less cooling and lower electricity costs in a lab). The newly assembled lab machines also benefit: M.2 NVMe drives eliminate SATA data and power cables, so fewer connections can be loose, directly reducing the 'drive not detected' errors the IT team reports. Reduced downtime and support calls outweigh the slightly higher cost per GB, and 256-512 GB is sufficient for office data (bulk data can be kept on a file server or cloud).

Section D — AI-Augmented Learning

Submission item 1: Exact prompt given to the AI tool

"Create a step-by-step troubleshooting checklist for a junior technician to diagnose a desktop PC that fails to boot immediately after assembly. Put the steps in the correct priority order. It must cover at least the motherboard, RAM, storage and PSU as possible failure points, be presented as a clear checklist or simple decision flow a technician can follow on-site, and then apply it to this scenario: ABC Solutions is setting up a new computer lab; after assembling several systems, some computers fail to start while others display hardware detection errors, and the team suspects incorrect installation or compatibility issues. Show which step would have caught the reported issue."

Submission item 2a: The AI's original checklist

- 1. Check BIOS/UEFI settings and update the BIOS to the latest version.
- 2. Check that the operating system is installed correctly and the boot order is set to the right drive.
- 3. Check the RAM: reseat the modules and try one stick at a time.
- 4. Check the storage drive: make sure the SATA/NVMe drive is connected and detected.
- 5. Check the PSU: confirm the 24-pin and CPU connectors are plugged in and the PSU switch is on.
- 6. Check the motherboard for visible damage, bent pins, or loose standoffs.
- 7. Reseat the CPU and the GPU.
- 8. If it still fails, replace components one by one.
- **Application to ABC Solutions:** Step 4 would catch the detection errors (drives not connected) and step 5 would catch the machines that fail to start.

Submission item 2b: Corrected version (my fixes marked CHANGED / ADDED)

Step	Corrected checklist (priority order)	Note
0	Safety first: power off, switch the PSU off, unplug the mains cable, hold the power button to discharge, put on an ESD wrist strap or touch the bare chassis, and work on an anti-static surface.	ADDED
1	External power and basics: wall socket/power strip works, mains cable seated, PSU switch on (I), PSU voltage selector correct, monitor cable in the correct output (GPU if installed), monitor on correct input.	ADDED / MOVED UP
2	PSU connections: 24-pin ATX and CPU 4/8-pin EPS fully clicked in, GPU PCIe power connected, front-panel power switch header on the correct pins. If no fans/lights at all, test the PSU (paperclip/PSU tester) or swap in a known-good PSU.	CHANGED (moved from step 5)
3	Visual inspection of the motherboard and case: no loose or extra standoffs shorting the board, no stray screws, no bent CPU socket pins, cooler fitted with film removed and CPU_FAN connected.	CHANGED (moved up)
4	Minimal boot: disconnect all non-essential items (extra drives, cards, USB devices) and boot with just CPU, cooler, one RAM stick, PSU (+GPU if no iGPU).	ADDED
5	RAM: reseat until clips click, try the recommended slot (A2), test one module at a time and a known-good module; check type (DDR4/DDR5) and speed.	Kept
6	CPU and board compatibility: confirm socket/chipset match, the BIOS version supports the CPU (use BIOS flashback if needed), reseat the CPU, clear CMOS.	CHANGED (BIOS update only after the hardware checks, and only when needed)

7	Storage: reseal and screw down the M.2 / check SATA data and power cables; verify the drive appears in BIOS; try another port/slot.	Kept
8	GPU: reseal, connect PCIe power, test on the iGPU or a known-good card.	Kept
Step	Corrected checklist (priority order)	Note
9	Software/boot settings (last): boot order, SATA/NVMe mode, OS installation/boot media.	CHANGED (moved from steps 1-2 to the end)
10	Isolate and replace: swap one known-good component at a time; document the results in the deployment log.	Kept

Decision flow summary: No power/fans? → go to steps 1-2 (power). Fans spin but no display/beeps? → steps 4-6 (RAM/CPU/board, minimal boot). POST OK but 'no boot device'/drive not found? → step 7 (storage).

Everything detected but fails to load an OS? → step 9 (software).

Applied to ABC Solutions: Machines that *fail to start* would be caught at **step 2** (unseated 24-pin/EPS connectors, front-panel power header) or step 3 (shorted standoff). Machines showing *hardware detection errors* would be caught at **step 5** (RAM not seated/incompatible), **step 6** (CPU not supported by the BIOS) and **step 7** (drive not seated or cable loose).

Submission item 3: Note on what I changed and why

The AI's list had no ESD/electrical safety step, so I added step 0 to protect both the technician and the components. It also began with BIOS updating and OS checks, which are software checks that cannot run if the machine has no power or fails POST, so I moved them to the end and put power and cable checks first, since they are the quickest and most common cause of failure. I added a minimal-boot step to isolate the faulty part and moved the BIOS update into the CPU compatibility step, since flashing BIOS blindly can be risky. The original also lacked basic checks (mains, monitor input, front-panel header), which I added as step 1.

Reminder: for Section D, submit the prompt you really used and the AI output you really received, and make the Step 2 review in your own words; adjust the text above to match.